

Software Requirements Specification

for

Student Project Management Information System (SPMIS)

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1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the Student Project Management Information System (SPMIS). It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli.

1.2 Document Conventions

This document follows the IEEE formatting requirements for SRS. The standard Arial font size 11 is used for standard text, and bold styling is used for key emphasis.

1.3 Intended Audience and Reading Suggestions

This document is intended for project stakeholders, developers, university administrators, supervisors, and students. Readers should start from the Overview and proceed to the functional and non-functional requirements.

1.4 Product Scope

SPMIS is a web-based application designed to manage and monitor academic projects in educational institutions. The system will allow project registration, proposal submission, task management, milestone tracking, weekly progress reporting, supervisor feedback, and final project submission. The system aims to improve transparency, accountability, and overall project management efficiency.

1.5 References

- Microsoft Documentation – ASP.NET Core Web API
- IEEE Software Engineering Standards
- UML Modeling Guidelines

2. Overall Description

2.1 Product Perspective

The system is a new self-contained web-based application that follows a three-tier architecture consisting of a Presentation Layer (Frontend), Application Layer (Backend API using ASP.NET Core), and a Data Layer (SQL Server).

2.2 Product Functions

- Project registration and group management
- Proposal submission and approval
- Task creation, assignment, and tracking
- Weekly progress reporting and milestone tracking
- Supervisor feedback and project evaluation
- Report generation and project archiving

2.3 User Classes and Characteristics

- Students: Users who will form groups, register projects, submit proposals, manage tasks, and upload project documents.
- Supervisors: Faculty members who will review proposals, assign tasks, provide feedback, and evaluate final submissions.
- Admins: Users who manage the system configuration, user roles, and monitor system analytics.

2.4 Operating Environment

The system is web-based. The frontend runs in modern web browsers (Chrome, Firefox, Safari). The backend will be hosted on a Windows Server running IIS to support ASP.NET Core Web API, and data will reside in a Microsoft SQL Server database.

3. External Interface Requirements

3.1 User Interfaces

The user interface will be built using HTML, CSS, JavaScript, and Bootstrap. It will be responsive to accommodate different screen sizes and will feature specific dashboards for Students and Supervisors.

3.2 Software Interfaces

The backend API will interface with the SQL Server database via Entity Framework Core. Secure RESTful endpoints will be provided for frontend consumption.

4. System Features

4.1 Project and Group Management

The system shall allow students to form groups and register their academic projects. It must validate group sizes and ensure unique project titles within a given academic session.

4.2 Task and Milestone Tracking

The system shall enable task creation, assignment to group members, and status tracking (e.g., To Do, In Progress, Done). It will also allow the setting of project milestones.

4.3 Feedback and Evaluation

The system shall provide a mechanism for supervisors to leave comments on weekly reports, approve deliverables, and submit final evaluation scores.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should handle at least 500 concurrent users without significant degradation in response time. Page load times should be under 2 seconds.

5.2 Security Requirements

User authentication and authorization will be required. Passwords must be hashed, and communication between client and server should be encrypted using HTTPS.

5.3 Software Quality Attributes

The system will emphasize usability, ensuring a simple and intuitive interface. Reliability is critical, and the database should be backed up regularly to prevent data loss.